A Project Proposal on

# SCHOOL MANAGEMENT SYSTEM

Submitted in Partial Fulfillment of the Requirement for Degree of Bachelor of Science in Computer Science and Information Technology (B.Sc. CSIT) Awarded by Tribhuvan University

**Submitted by:**

**Manish Pandey (79011946)**

**Priya Shrestha (79011949)**

**Sabina B.k (79011955)**

**Supervisor:**

**Er. /Mr. Supervisor Name**

**Designation**

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

**MOUNT ANNAPURNA CAMPUS**

**Pokhara-5, Parshyang**

**February 2026**

A Project Proposal on

# SCHOOL MANAGEMENT SYSTEM


**Submitted by:**

**Manish Pandey (79011946)**

**Priya Shrestha (79011949)**

**Sabina B.k (79011955)**

**Supervisor:**

**Er. /Mr. Supervisor Name**

**Designation**

**Submitted to:**

**DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY**

**Mount Annapurna Campus**


**February 2026**

## STUDENT'S DECLARATION

We hereby declare that this project entitled **School Management System** is based on our original work. All concepts, data, code, and any other work from external sources have been properly cited and referenced in accordance with the guidelines provided by Institute of Science and Technology, Tribhuvan University.

We owe all the liabilities relating to the authenticity and originality of this project work and project report.

I. Manish Pandey (79011946) ……………………………

II. Priya Shrestha (79011949) ……………………………

III. Sabina B.k (79011955) ……………………………

Date: ……………………….

# Department of Computer Science and Information Technology

# MOUNT ANNAPURNA CAMPUS

## APPROVAL LETTER

The undersigned certified that the project entitled **School Management System** submitted by **Manish Pandey, Priya Shrestha, Sabina B.K** for partial fulfillment of the degree of Bachelor of Science in Computer Science and Information Technology has been officially approved to undertake this as an academic project.

We extend our best wishes for the successful execution and completion of the project

…………………………..

**Internal Examiner Name**

Internal Examiner

Department of CSIT

Mount Annapurna Campus

…………………………..

**Supervisor Name**

Supervisor

Department of CSIT

Mount Annapurna Campus

…………………………..

**Director Name**

Director

Department of CSIT

Mount Annapurna Campus

Date: ……………………….

# ABSTRACT

Most educational institutions in Nepal continue to rely on manual registers, paper-based notes, and disconnected processes for recoding attendance, distributing assignments and class materials, and sharing examination results. These traditional methods lead to significant inefficiencies, including time wasted during roll calls, frequent occurrences of proxy attendance, loss or misplacement study materials, delayed dissemination of exam results, and limited communication between schools, teachers, students, and parents.This project develops a web-based School Management System designed to address these challenges through a centralized, secure, and user-friendly digital platform. The system incorporates three primary features: geofencing-enabled attendance management ,which allows teachers to mark attendance only when physically present within a 20-meter radius of the school building using browser-based GPS verification; assignment and class notes upload functionality ,enabling teachers to upload PDF documents that students and parents can easily view and download ;and exam report management ,which permits teachers to securely upload results and progress reports for students and parents to access in real time. Built using modern web teachnologies-Next.js (App Router) with TypeScript for full-Stack development, Tailwind CSS and Shadcn/UI for responsive styling, NextAuth with JWT for secure authentication, and PostgreSQL with Prisma ORM for efficient database management- the system provides role-based access control for administrators, teachers, students, and parents. It supports real-time attendance tracking, automated notifications for low attendance, new assignments, or published exam reports, and intuitive dashboards for all users. The primary goal of the system is to modernize daily school operations in Nepal by eliminating proxy attendance, streamlining the distribution and access of learning materials, reducing administrative workload, improving record accuracy and availability, enhancing transparency in academic progress, and strengthening communication among all stakeholders.

**Keywords:** *Geofencing, Attendance, Assignment Upload, Exam Report, Digital Education, Role-Based Access Control*

# TABLE OF CONTENTS

# LIST OF ABBREVIATIONS

| | |
|---|---|
| API | APPLICATION PROGRAMMING INTERFACE |
| GPS | GLOBAL POSITIONING SYSTEM |
| JWT | JSON WEB TOKEN |
| ORM | OBJECT RELATIONAL MAPPING |
| PDF | PORTABLE DOCUMENT FORMAT |
| RBAC | ROLE-BASED ACCESS CONTROL |
| UI | USER INTERFACE |

# LIST OF FIGURES

# LIST OF TABLES

# CHAPTER 1
# INTRODUCTION

## 1.1 Background

Running a school well is much harder than most people think. Every day, teachers, and administrative have to handle many important tasks: taking accurate attendance so no one is marked present when they're absent, sharing study materials like assignments, class notes, and handouts so every student gets what they need, giving out exam results and progress reports clearly and on time ,and keeping good communication going between teachers, students, parents, and progress reports clearly and on time ,and keeping good communication going between teachers, students, parents, and the school office .When these things are done properly ,students stay more responsible ,learn better ,parents feel more connected and can help at home ,problems like frequent absence or falling grades get notices early, and the while school runs more smoothly [1] [2].

Unfortunately, in most schools across Nepal –whether government, community, or small private's ones-these basic tasks are still done the old-fashioned way. Teachers stand at the front of the class calling out names from a paper register or quickly typing names into a shared Excel file on one computer. This simple roll call often takes 5-10 minutes of class time, and it's easy for mistakes to happen: names get skipped written wrong, or-worst of all-one student says "present" for a friend who didn't come [2] [3].Assignments and class notes are usually handed out on paper, written on the blackboard for everyone to copy, or sometimes sent as photos through WhatsApp or Viber. If a student misses' class or loses the paper, they miss important work, and the teacher must explain everything again later. This wastes time and makes it hard for everyone to stay on the same page [4] [5]. Exam results and progress reports face the same problem. Marks are often just announced in class, scribbled on a notice board, or sent a quick message to parents' phones. This means results can arrive late, parents misunderstand the marks or forget them, there's no proper record to look back on later, and students don't get consistent feedback about how they're doing over time [6] [7].

All these old methods create real difficulties. Teachers spend too much time on paperwork instead of teaching. Students don't always get the materials or results they need when they need them. Parents stay mostly in the dark, about what's happening at school every day, so they can't help supporting their child properly.

These problems are especially tough in Nepal. Many schools have very little money to spend on technology ,electricity and internet that often stop working ,not enough computers or trained people to manage them, and a long habit doing everything on paper [4] [8].Some private schools unities use paid software, but it's usually expensive ,needs mobile app, doesn't work well in villages or areas with poor internet [9] [10].

Thankfully, today's web technologies make it much easier to build good digital tools. Modern frameworks like NEXT.js let us create fast, nice –looking websites that work on any browser-whether on a school desktop, a teacher's laptop, or parents' phone. Secure login systems, simple databases, and easy file handling mean we can make a complete system without needing expensive apps or special equipment [11] [12].

Our School Management System is designed exactly for this situation. It brings everything into one easy-to-use website:

- Attendance that only works when the teacher is really inside the school (using the phone's GPS to check they're within 20 meters).
- Simple upload of assignments and notes as PDFs so students and parents can always see and download them.
- Secure upload and view exam results and reports, so everyone knows the marks right away.

This system cuts out fake attendance, saves paper, makes sure materials and results reach everyone on time, and helps teachers, students and parents stay better connected-all while being affordable and realistic for schools across Nepal.

## 1.2 Problem Statement

Manual and fragmented school processes cause inefficiency, proxy of attendance, loss of important study materials, delayed exam results of dissemination, and weak communication between school and parents. Existing digital solutions are often expensive, complex, or lack strong attendance verification. There is a need for an affordable, web-based school management System that combines secure attendance, document upload, and result management [1] [11].

The core problems observed in most Nepalese educational institutions can be summarized as follows:

- Manual Attendance process wastes time and area prone to errors.
- Proxy attendance is common and hard to prevent.
- Study materials (assignments /notes) are often lost or unequally distributed.
- Exam results are shared late and informally.
- Poor Communications between school and parents.
- Existing digital solutions are expensive or too complex.
- Most tools lack strong attendance verification.
- Nepal's schools face budget, internet, and technical limitations.

## 1.3 Objective

The main aim of this project is to develop a modern web-based School Management System for Nepalese educational institutions. The specific objectives are:

- To create a centralized platform with geofencing enabled attendance that restricts marking to within 20 meters of school premises.
- To provide role0-based access, assignment/note upload (PDF), exam report management, real-time data, and notifications to improve efficiency, accuracy, and communication.

## 1.4 Features

The Education Management System offers a comprehensive set of features designed to streamline academic and administrative processes while ensuring security and accessibility for all users. It enables secure access through role-based authentication, simplifies student and teacher management, and provides real-time insights and notifications to enhance attendance tracking and overall academic monitoring.

Key features include:

- **Secure authentication and role-based access:** To control ensures that only authorized users can access their respective dashboards.
- **Students, teachers, and class management:** To allow efficient organization and management of academic data.
- **GPS geofencing (20 m radius) for attendance marking:** To ensure accurate attendance records by verifying location.
- **Teacher attendance marking interface**: To provides an easy–to-use portal for teachers to marked attendance
- **Student/parent attendance viewing portal-** To allow students and parents to monitor attendance records.
- **Admin real-time analytics dashboard-** To present key metrices and insights for informed decision-marking.

## 1.5 Applications

The Educational Management system is suitable for a variety of educational institutions that aim. To improve academic management, attendance tracking, and communication between stakeholders. Its flexibility makes it ideal for both government and private schools, as well as colleges looking to implement digital monitoring and reporting systems.

Key applications include:

- **Government and private schools/colleges in Nepal:** To streamlines administration and academic operations.
- **Institutional needing strict attendance monitoring:** To ensure accurate attendance tracking using GPS geofencing.
- **Parent-teacher communication platforms:** To facilities transparent communication regarding student performance and attendance.

# CHAPTER 2
# LITERATURE REVIEW

## 2.1 Background Study

Attendance tracking is essential but time-consuming in educational institutions. In Nepal, Manual registers and spreadsheets remain common. Leading to inefficiencies, human errors, proxy attendance, and poor data management [1] [2] [8]. Digital solutions such as RFID, barcode scanning, and biometric systems have been introduced, but they require expensive hardware and ongoing maintenance, limiting adoption in resource-constrained Nepalese schools [13] [14] [3].

Geofencing provides a low-cost alternative by using GPS to define a virtual perimeter. Only users within the specified radius can mark attendance, significantly reducing proxy cases [3][8][18]. Browser-based geofencing through the HTML5 Geolocation API enables simple web deployment – well-suited to low-resource stings [9] [10].

Digital platforms for assignment distribution and exam report sharing have also shown benefits in accessibility and reduced paper usage when implemented securely [4] [3].

## 2.2 Review of Similar Projects

Numerous studies and practical implementations have explored various aspects of school management systems, particularly automated attendance tracking, digital resource distribution, and academic reporting. These works offer valuable insights into technological approaches, their effectiveness, limitations, and applicability to resource-constrained environments like those in Nepal.

Several research efforts have focused specifically on geofencing-based attendance solutions. Ewe Oya et al. designed and implemented a university-level attendance management system that utilized mobile geofencing to restrict attendance marking to designated campus zones [15]. Their evaluation showed a significant reduction in proxy attendance and improved overall administrative efficiency. Similarly, Soko and Charola develop a geofencing attendance tracker that achieved comparable results in minimizing unauthorized entries [16]. However, both systems were built around native mobile applications and employed relatively wide geofence radii (typically 100-500 meters), making them more suitable for large university campuses then the compact school buildings commonly found in Nepal [17]. The reliance on mobile apps also introduces challenges related to device compatibility, installation requirements and maintenance overhead, which are often impractical in Nepalese government and small private schools. More recent studies have shifted towards web-based geofencing solutions that leverage the HTML5 Geolocation API. Research published in IEEE conference proceedings [11] presented a fully browser-based geofencing attendance system, emphasizing advantages such as no need for app installation, reduced development and deployment costs, and broader accessibility across devices. a similar web-oriented approach was documented in IJASET [13], where the authors highlighted its suitability for low resource educational settings, these browser-based systems are particularly relevant for Nepal [18] [12].

Commercial platforms have also contributed to the field. Solutions like Sweedu Edu tech [9] and Katmatic [10] provide integrated school management tools that combine geofencing attendance, document/files sharing capabilities, and parent communication portals. These products demonstrate real-world usability, including real-time notifications, multi-user dashboards, and basic file upload features. However, their subscription-based pricing model often makes they financially unviable for the government schools/collage and small private institutions on Nepal, where budget limitations are a major constraint [4]. Beyond attendance, other research has investigated digital management of assignments, notes, and examination reporting. Studies on web-based assignment distribution systems [5] and secure exam result portals [6] have shown that allowing teachers to upload documents (such as PDFs) and enabling students/parents to access them digitally reduces paper dependency, minimizes material loss, and accelerates feedback cycles. Additional works have evaluated the broader impact of digital tools on school administration, reporting improvements in transparency, parent engagement, student accountability, and overall academic performance when attendance, learning materials and results are managed through a unified digital platform [3] [7].

Despite these advancements, several persistent limitations remain across the reviewed literature. Most existing solutions either concentrate narrowly on attendance tracking without integrating document sharing or result management, depend heavily on mobile-only applications, or adopt wide geofence boundaries that are impractical for smaller school premises. Moreover, very few systems are specifically designed, tested, or optimized for Nepal’s unique challenges – including intermittent internet connectivity, variable GPS signal quality (especially indoors), limited technical support, and severe budget constraints [1] [8].

# CHAPTER 3
# METHODOLOGY

## 3.1 Requirement Analysis

The Education Management System is designed to address the academic and administrative needs of schools and colleges while ensuring accurate attendance tracking and secure access for all users.

### 3.1.1 Functional Requirements

The Education Management System is designed to support key academic and administrative operations efficiently. The core functions focus on secure access, accurate attendance tracking, and easy monitoring for all users. By combining location-based verification with robust reporting features, the system ensures accountability and transparency in daily school activities.

Keys functional features include:

- **User authentication and role management:** To secure login and role-based access for admins, teachers, students, and parents.
- **Geofencing verification before attendance marking:** To ensure attendance is recorded only within the permitted location radius
- **Attendance of recording, viewing, and reporting:** To allows teachers to mark attendance and students/parents to view records, with reports generated for administrative purposes.

### 3.1.2 Non-Functional Requirements

In addition to functional features, the Education Management System emphasizes quality attributes that ensure the system is reliable, secure, and user-friendly. These non-functional requirements focus on providing a seam; less experience for all users while safeguarding sensitive academic and personal data. High performance, consistent reliability, and strong data privacy measures are key aspects that the system is practical and trustworthy for schools and colleges.

Key non-functional features include:

- Usability- intuitive interfaces designed for admin, teachers, students, and parents.
- Security protection of sensitive academic and personal data against unauthorized access.
- Performance smooth and responsive operation with minimal delays.
- Reliability-consistent, accurate attendance tracking, and reporting.
- Data privacy compliances with privacy standards to protect user location and personal information.

## 3.2 Feasibility Analysis

To further justify the practicality of the proposed School Management System, additional explanations for each feasible in the ability aspect are provided below.

### 3.2.1 Technical Feasibility

The proposed system is technically feasible using modern, open-source, and widely supported technologies.

- **Frontend & backend:** Next.js (React framework with App Router) provides excellent performance, server-side rendering, and API routes.
- **Styling:** Tailwind CSS + Shadcn/UI ensures fast, responsive, and modern UI development.
- **Authentication:** NextAuth.js with JWT offers secure and simple role-based access.
- **Database:** PostgreSQL is robust, open-source, and works perfectly with Prisma ORM for type-safe database operations.
- **Geofencing:** HTML5 Geolocation API is natively supported in modern browsers and requires no extra libraries.
- **File handling:** PDF upload/download is straightforward with browser file API and server-side storage.

All required technologies are mature, free to use, and well-documented, with large community support. No advanced hardware or proprietary software is needed.

*Table 3.2.1.1: Hardware Requirement*

| **Component** | **Requirements** |
|---|---|
| Processor | Quad-Core (Intel i5 or higher) |
| RAM | 8GB |
| Storage | 128 GB SSD or higher |
| Internet | Stable broadband connection |
| Client Device | Laptop |

*Table 3.2.1.2: Software Requirement*

| Component | Technology/Version |
|---|---|
| Frontend Framework | Next.js (React) |
| Styling | Tailwind CSS + Shadcn /UI |
| Backend | Next.js API Routes |
| Authentication | NextAuth.js +JWT |
| Database | PostgreSQL |
| ORM | Prisma |
| Browser Support | Chrome, Fire Fox, Edge (latest version) |
| Operating System | Windows/macOS/Linux |
| Development Tools | Vs Code or similar IDE |
| Version Control | Git & GitHub |

### 3.2.2 Operational Feasibility

The system is highly operationally feasible:

- It runs entirely in the browser – no mobile app installation or special hardware is required.
- Teachers only need a smartphone/laptop with internet and GPS (most devices already have this).
- Minimal training is needed: Interface is intuitive and like common web applications.
- The school can use a single laptop/computer as the admin station.
- Parents and students access it via any smartphone browser.
- Ongoing maintenance is low (update via hosting platform).

The system aligns well with current school operations and reduces manual workload rather than disrupting it.

### 3.2.3 Economical Feasibility

The project is economically viable-source technologies, eliminating the need for costly software licenses.

- All technologies (Next.js, Tailwind, Prisma, PostgreSQL, NextAuth) are open-source and free.
- No licensing costs are involved.
- Hosting can be done on affordable platforms.
- Development is done by students-no external developers at cost.
- Long-term benefits include reduced paper usage, faster administrative processes, and better attendance monitoring, which outweigh minimal hosting expenses.

Thus, the system is affordable even for government and small private schools in Nepal.

### 3.2.4 Schedule Feasibility

The project is planned to use Agile methodology with parallel development of frontend and backend. The schedule below is realistic for a team of three final-year students and ensures completion within the academic timeline.

| Task Name | W 1 | W 2 | W 3 | W 4 | W 5 | W 6 | W 7 | W 8 | W 9 | W 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Requirement Analysis & Planning | | | | | | | | | | |
| System Design & Database | | | | | | | | | | |
| User Authentication & RBAC | | | | | | | | | | |
| Geofencing/ Attendance Module | | | | | | | | | | |
| Assignment/Notes Upload | | | | | | | | | | |
| Exam Report Upload & Viewing | | | | | | | | | | |
| Dashboard (Admin, Teacher, Student, Parent) | | | | | | | | | | |
| Notifications System | | | | | | | | | | |
| System Integration & API Testing | | | | | | | | | | |
| System Testing | | | | | | | | | | |
| Final Testing & Bug Fixing | | | | | | | | | | |
| Documentation | | | | | | | | | | |
| Report Writing | | | | | | | | | | |

Figure 3.2.4.1: Gannt Chart for School Management System

## 3.3 System Design

The School Management System follows a modern, full-stack web application architecture designed for simplicity, security, performance, and ease of maintenance. It is built as a single-page application (SPA) with server–side capabilities, making it responsive, fast, and suitable for low-resource environments like Nepalese schools.

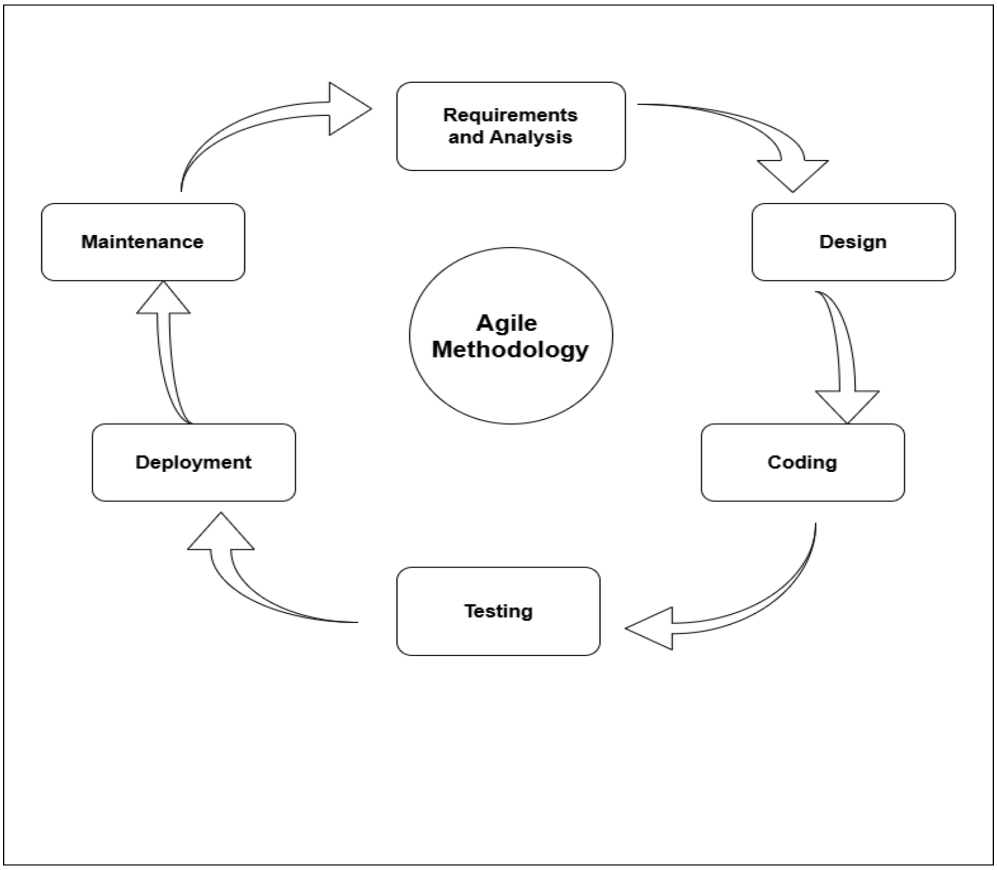


Figure 3.3.1:  Agile Sprint Development

### 3.3.1 Development Methodology

The project follows the Agile development methodology with the Scrum framework to build the School Management System in a flexible, collaborative, and efficient manner. Agile is chosen because it allows the team to adapt to change easily, detect problems early, deliver working features quickly, and work closely together as a small group of three students. This approach suits the academic timeline and helps us create a high-quality system step by step.

We work in 2-week sprints, where each sprint has a clear goal and delivers a usable part of the system. For example, early sprints focus on login, user roles, and database setup, while later sprints add geofencing attendance, assignment upload, exam reports, and final polish. At the start of each sprint, we plan tasks and decide what to complete. After every sprint, we review the completed work, show it to the supervisor or peers and hold a short retrospective to discuss what went well and what we can improve next time.

Every day, the team holds a quick stand–up meeting (10-15 minutes) to share progress plans, and any difficulties. This keeps everyone on the same page and helps solve issues fast. We use Git (GitHub) for version control-each feature developed in its own branch, reviewed through pull requests, and merged only after checking. Continuous integration is done automatically so that every change is tested quickly.

Key practices we follow:

- 2-week sprints with clear goals
- Daily stand-up meetings for quick updates
- Continuous testing and integration after every change
- Sprint reviews to show progress and get feedback
- Retrospectives to improve our process.
- GitHub for code sharing branching, and collaboration

### 3.3.2 Algorithm

The system implements several key algorithms to handle its core functionalities securely and efficiently. Below are the detailed algorithms for primary operations.

**A. Geofencing Attendance Marking**

**Purpose:** Allow attendance marking only when the teacher is inside the school (within 20 meters).

**Steps:**

Step 1. Get school location (latitude, longitude) from database.

Step 2. Ask browser for teacher's current location (using HTML% Geolocation API)

Step 3. If location access denied > show error: "Allow location to mark attendance".

Step 4. Calculate distance using Haversine formula:

- Distance = $2 \times 6371 \times \arcsin(\sqrt{(\sin^2((lat2-lat1)/2) + \cos(lat1) \times \cos(lat2) \times \sin^2((lon2-lon1)/2)})$ (in km)

Step 5. If distance <=0.02 km → allow marking.

Step 6. Else → show error: "you are not inside school premises".

Step 7. Save attendance: student ID, class ID, date, status (Present/Absent), marked by teacher, timestamp.

### B. Assignment / Notes Upload

**Purpose:** Let teachers upload PDF files for students/parents to view.

**Steps:**

Step 1. Check teacher is logged in and assigned to the class.

Step 2. Accept PDF file from teacher.

Step 3. Check: must be PDF and size <10MB.

Step 4. Create unique name

Step 5. Save file to the server folder.

Step 6. Save details in database: file name, path, class ID, upload date, title.

Step 7. Show success: “File uploaded successfully”.

### C. Exam Report Upload & Viewing

**Purpose:** Teachers upload exam reports; students/parents can see them.

**Upload Steps:**

Step 1. Verify the teacher and class.

Step 2. Accept the PDF report file.

Step 3. Validate: PDF only, size<10MB.

Step 4. Save file with unique name to server.

Step 5. Save report in database: report title, file path, class/student ID, teacher ID, date.

Step 6. Notify student/parent: “New exam reports available”.

**View Steps:**

Step 1. User logs in (students or parents).

Step 2. Check if the user has access to this report (own student or child).

Step 3. Show secure download/ view link.

Step 4. Log who viewed it.

## 3.4 Block Diagram

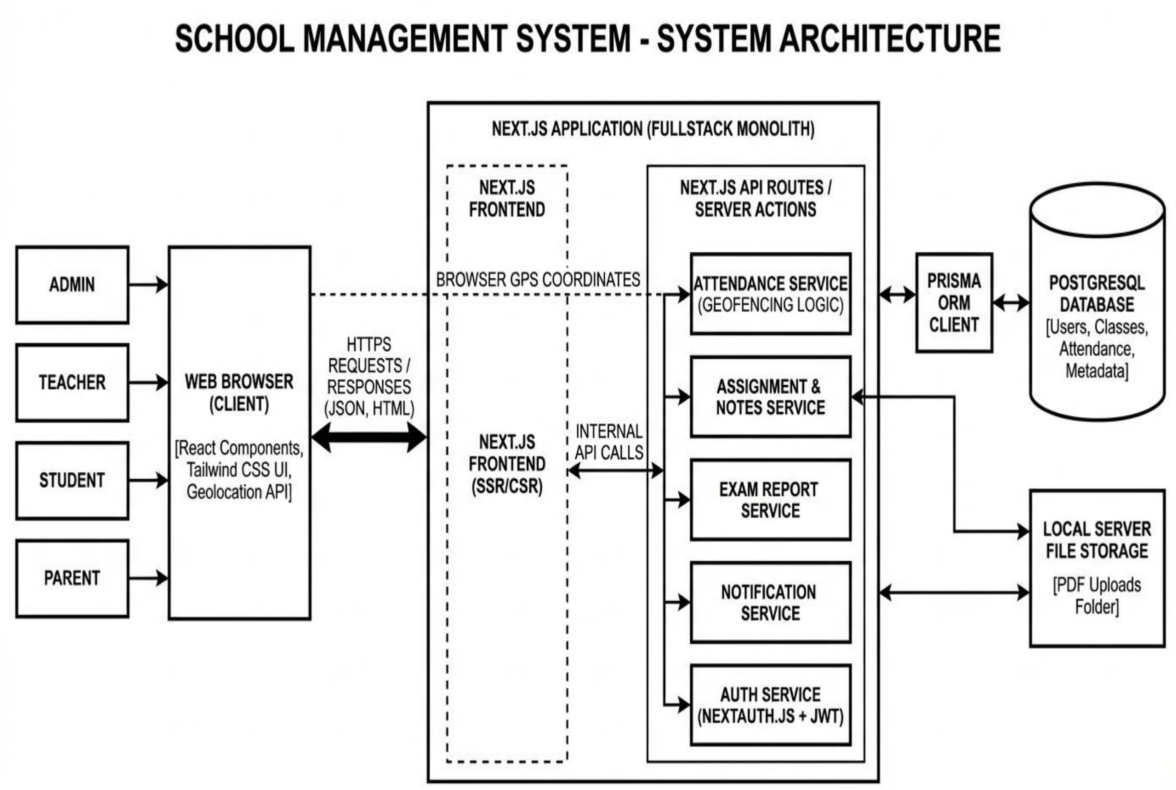


Figure 3.4.1: Block Diagram of School Management System

## 3.5 Flowchart

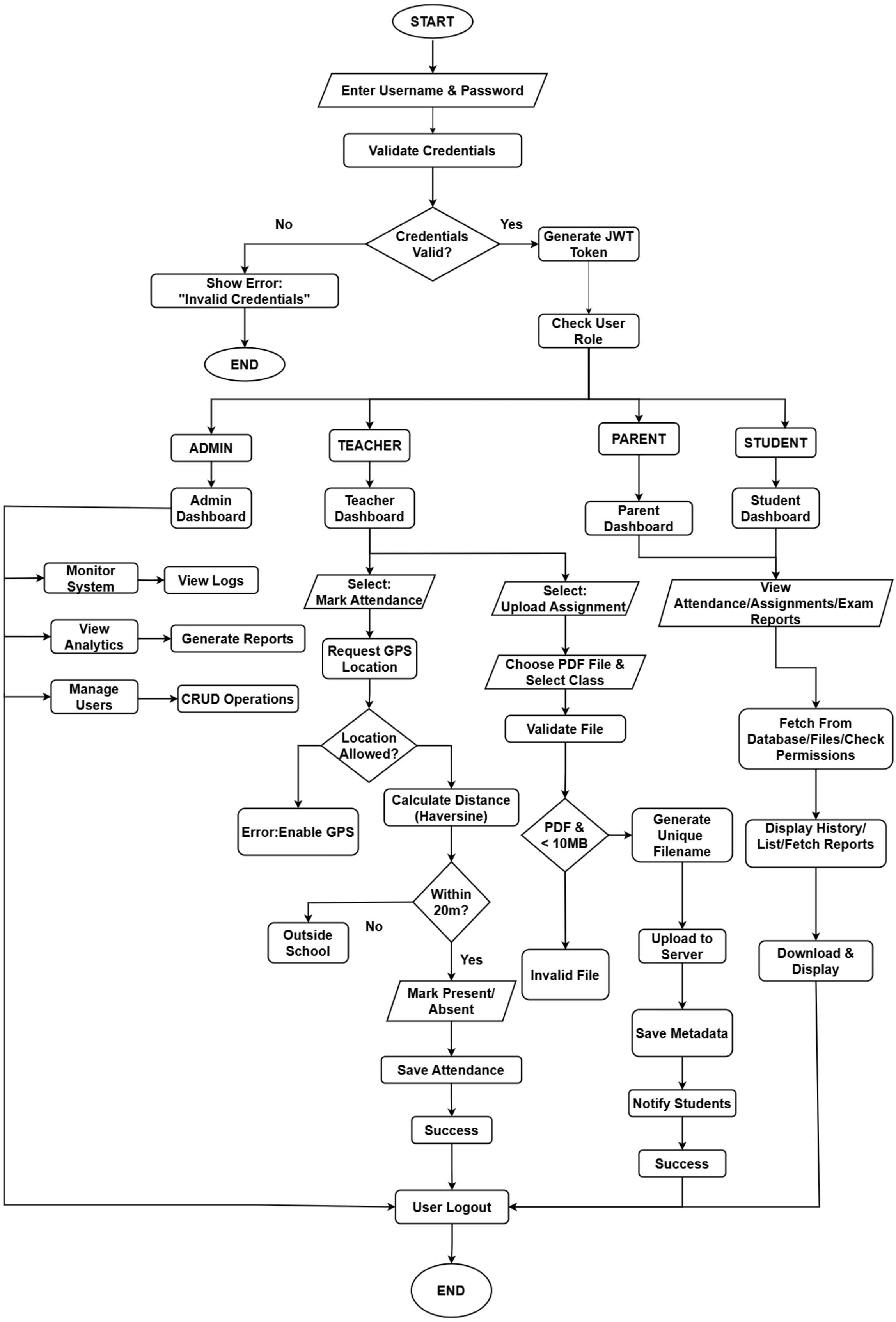


Figure 3.5.1: Flowchart of School Management System

## 3.6 Use case Diagram

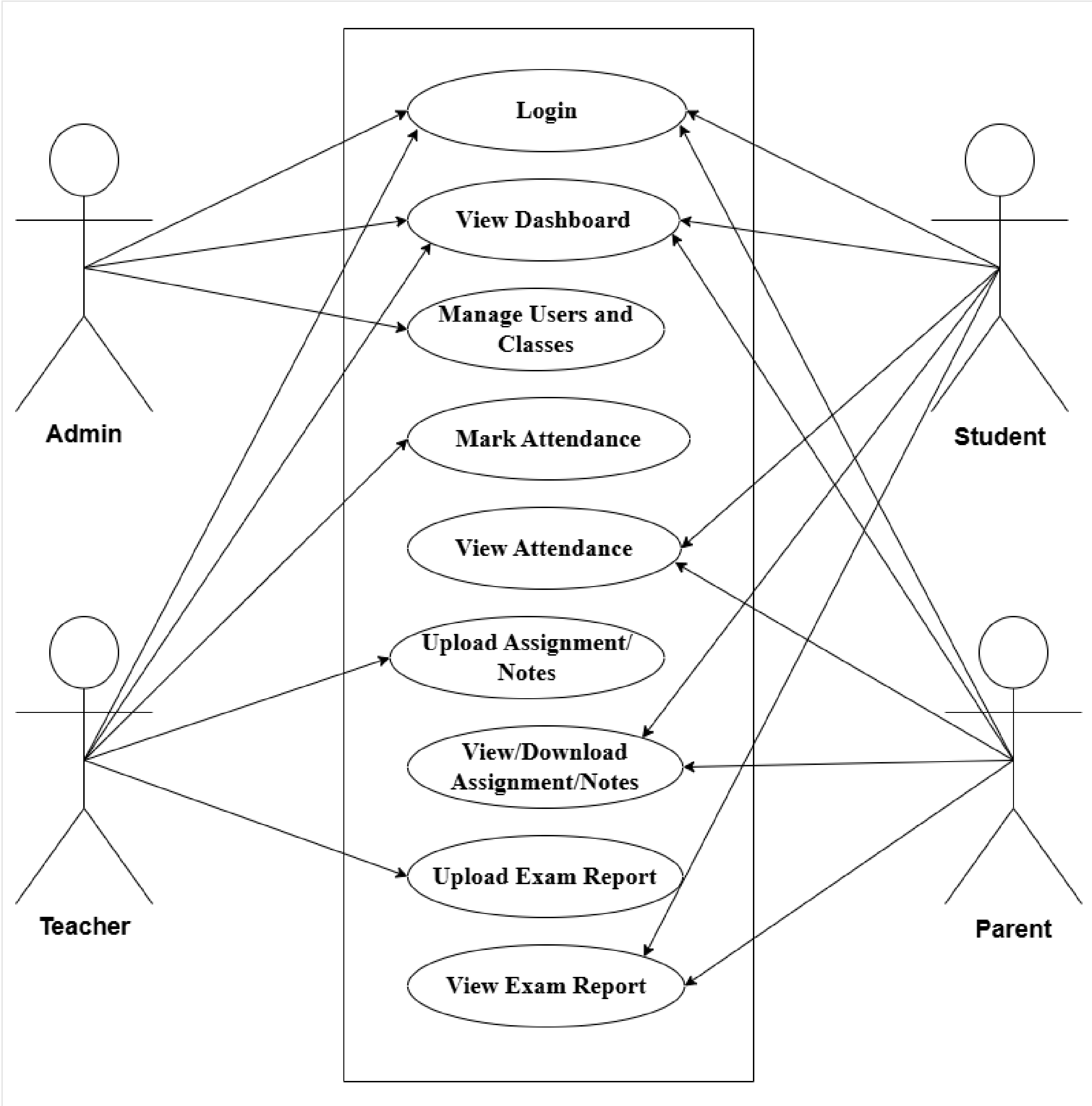


Figure 3.6.1: Use Case diagram of School Management System

# Chapter 4
# EXPECTED OUTCOME

Upon Successful implementation, the School Management System is expected to provide a centralized, secure, and user-friendly digital platform that significantly modernizes school administration in Nepal. The geofencing attendance feature will deliver accurate, tamper-resistant records by restricting marking to within a 20-meter radius of the school building, effectively eliminating proxy attendance and reducing manual roll-call time.

The assignment/notes upload module will allow teachers to share PDF documents (assignments, class notes, study materials) instantly-students can access and download them anytime, reducing paper usage and ensuring no student misses important materials.

The exam report management feature will enable teachers to upload results and progress reports securely- students and parents can view them in real-time, improving transparency and allowing timely follow-up on academic performances.

Administrators will benefit from real-time dashboards showing attendance trends, uploaded files overview, exam report status, and user activity. Teachers will manage attendance, materials, and results efficiently. Students and parents will stay informed through attendance history, downloaded assignments, exam reports, and automated notifications (low attendance, new assignments, results published).

Overall, the system is anticipated to streamline administrative tasks, enhance data accuracy and accessibility, reduce paperwork, improve communication among all stakeholders, support better monitoring of student performance and regularity, and contribute to a more organized, transparent, and efficient academic environment in Newplases educational institutions.

# REFERENCES


[1] N. Bhattarai, R. Karki and S. Adhikari, "Factors Affecting School Attendance and Implications for Student Achievement by Gender in Nepal," *Review of Political Economy,* vol. 32, no. 4, p. 567–589, 2020.

[2] Orah, "The Hidden Costs: How Manual Attendance Tracking Damages Schools," Orah Blog, 2024. [Online]. Available: https://www.orah.com/blog/manual-attendance-costs. [Accessed feb 2026].

[3] "Impact of Digital Attendance System on Student Performance and Discipline," *International Advanced Research Journal in Science, Engineering and Technology (IARJSET),* vol. 12, no. 1, p. 67–74, 2025.

[4] R. Shrestha, P. Thapa and K. Gurung, "Digital Transformation in Nepalese Education: Opportunities and Challenges," *Journal of Information Technology Nepal,* vol. 7, no. 1, p. 22–35, 2024.

[5] M. Singh and P. Gupta, "Web-Based Geofencing Attendance System for Resource-Constrained Institutions," *International Journal of Engineering Research and Technology,* vol. 14, no. 3, p. 89–97, 2025.

[6] "Geofencing for Attendance: Top 11 Advantages," Workstatus, 2023. [Online]. Available: https://www.workstatus.io/blog/geofencing-attendance-advantages. [Accessed feb 2026].

[7] "Enhancing Attendance Tracking Efficiency through Location-Based Systems," *RSIS International Journal of Research in Innovation and Social Science,* vol. 8, no. 4, p. 112–120, 2024.

[8] B. Acharya, "Challenges of MIS Implementation in Nepalese Educational Institutions," *Nepal Journal of Science and Technology,* vol. 18, no. 2, p. 45–52, 2019.

[9] S. Edutech, "Geofencing Attendance System: The Future of Smart Attendance," 2025. [Online]. Available: https://sweedu.com/geofencing-attendance. [Accessed feb 2026].

[10] "Smart School Attendance System | Transforming Education in Nepal," Katmatic, 2025. [Online]. Available: https://katmatic.com/smart-attendance-nepal. [Accessed feb 2026].

[11] "Attendance Management System Using Geofencing Technology," IEEE, 2024.

[12] S. Patel and R. Sharma, "Hybrid Attendance System with Geofencing and Biometric Verification," *International Conference on Smart Computing and Communication,* p. 156–163, 2024.

[13] "Tick Mark: Geofencing Attendance System," *International Journal of Research in Applied Science and Engineering Technology (IJRASET),* vol. 13, no. 2, p. 234–241, 2025.

[14] "TECH-CLASS: A Geofencing-Driven Attendance Management System for Higher Education," *Journal of Mathematics and Computer Science,* vol. 15, no. 3, p. 78–89, 2025.

[15] I. Eweoya, A. Bello and T. Adebayo, "Design and Implementation of a University Attendance Management System Using Geo-Fencing," *Asian Journal of Computer Science and Technology,* vol. 14, no. 1, p. 45–56, 2025.

[16] H. Soko and F. Chatola, "Attendance Tracking System using Geofencing," *i-manager's Journal of Educational Technology,* vol. 21, no. 4, p. 34–42, 2024.

[17] "Mobile Based Student Attendance System Using Geo-Fencing With Timing and Face Recognition," ResearchGate, 2022. [Online]. Available: https://www.researchgate.net/publication/360123456. [Accessed feb 2026].

[18] A. Kumar, S. Verma and P. Singh, "Location-Based Attendance Monitoring Using Geofencing and Machine Learning," *International Journal of Advanced Computer Science and Applications,* vol. 16, no. 5, p. 201–210, 2025.